

Chapter 1: Why Does Interpretation Diverge?

We interpret the words of others every day.

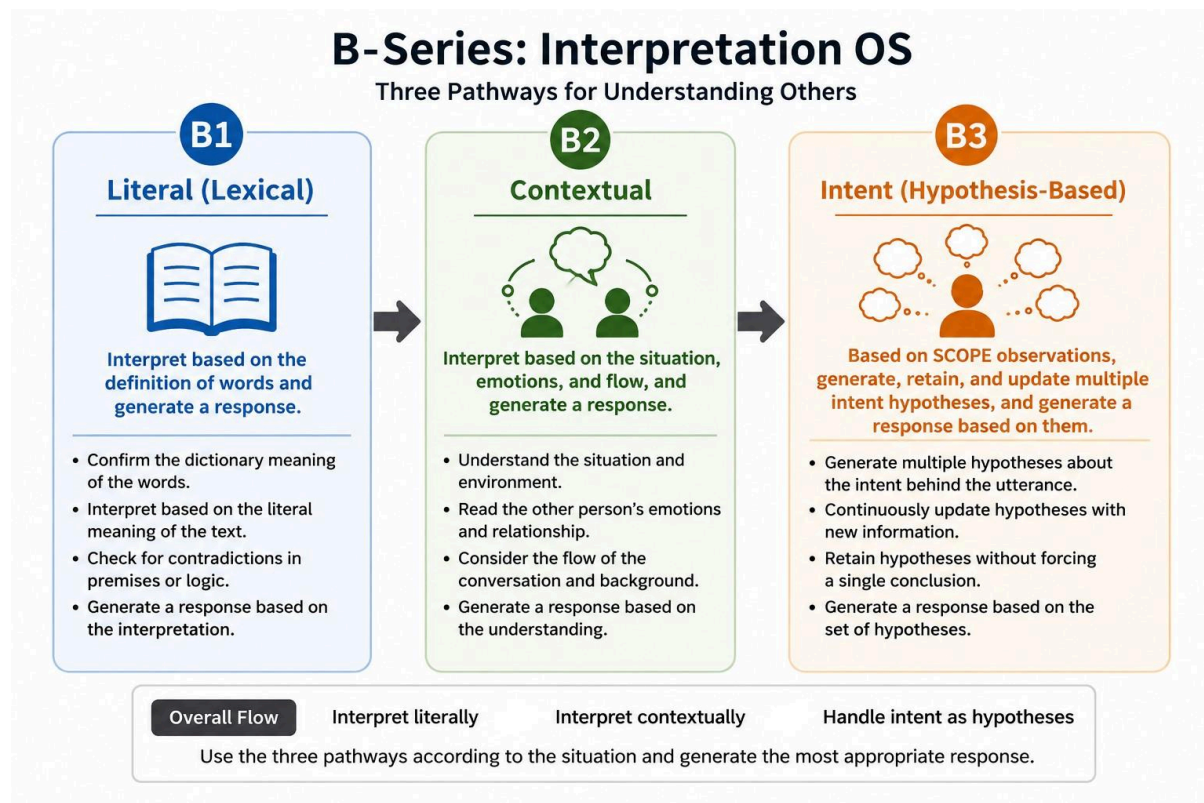
Consider the phrase "Keep it up!" The responses it generates are far from uniform. Some receive it as straightforward encouragement. Others hear it as criticism — "I'm already doing everything I can." Still others experience it as pressure. The speaker is one person, and the words are the same. Yet the interpretation is not necessarily singular.

This phenomenon is not limited to human-to-human communication. Even when an output is designed to be received as encouragement, a divergence in the user's interpretation may prevent the exchange from functioning as intended. The same dynamic has been observed in customer support contexts. An interaction designed as a considerate response may be received by the customer as something else entirely.

Divergence in interpretation may not be a matter of whether the utterance was appropriate or not. It may be a structural problem.

The B-Series is a model that addresses this divergence in interpretation by organizing the process from interpretation through to response generation into three distinct pathways.

Chapter 2: What Is the B-Series?



The B-Series is the interpretation and response generation module within COS (Cognitive Operating System).

Its role is not to produce the correct interpretation. It is to organize the pathways through which the utterances of others are interpreted and responses are generated, and to make those pathways observable.

Interpretation is often perceived as a single act. In practice, however, the pathway through which interpretation occurs may not be singular.

The B-Series organizes these pathways into three categories.

- B1: Literal
- B2: Contextual
- B3: Intentional

These three pathways do not represent levels of proficiency or a hierarchy of quality. Nor does any single pathway apply uniformly to all utterances in all situations.

What has been observed, however, is that the pathway most readily referenced tends to vary depending on the individual and the situation.

The purpose of the B-Series is not to guarantee correct interpretation. It is to make the process from interpretation through to response generation observable as a structure.

Chapter 3: B1 — Literal

B1 is the pathway that interprets based on the definition of words and generates a response accordingly.

In B1, the dictionary meaning of the words contained in an utterance is confirmed, and the utterance is interpreted according to its literal meaning. From there, logical consistency is verified, and a response is generated based on that result.

In this pathway, the definition of the words themselves takes priority as the primary reference point over situational context or relational factors.

B1 tends to function effectively in situations where the literal meaning of an utterance is clear. On the other hand, when information not contained in the utterance itself is important to its interpretation, B1 alone may not capture the full picture of what is being communicated.

The purpose of B1 is not to guarantee correct interpretation. It is to make the process from interpretation through to response generation observable, using the definition of words as its basis.

[Example] Upon receiving the utterance "Keep it up!", B1 interprets it literally as either encouragement or an instruction. The resulting response is either an assessment of whether further effort is possible, or an expression of gratitude for the encouragement.

Chapter 4: B2 — Contextual

B2 is the pathway that interprets based on situation, emotion, and flow, and generates a response accordingly.

In B2, processing proceeds based not on the utterance itself, but on how the receiver took in that utterance and what kind of response would be most appropriate. The atmosphere of the situation, emotional tone, and relational context serve as the primary reference points.

In this pathway, the basis for interpretation tends to be placed in the receiver's own emotional experience and relational context.

B2 tends to function effectively in situations where emotion and relational dynamics play a significant role. On the other hand, because the basis for interpretation lies within the receiver, there is a possibility that the interpretation may diverge from the speaker's intent.

The purpose of B2 is not to guarantee correct interpretation. It is to make the process from interpretation through to response generation observable, using emotion, situation, and relational context as its basis.

[Example] Upon receiving the utterance "Keep it up!", B2 proceeds based on how the receiver felt upon hearing it. If received as encouragement, gratitude is returned. If experienced as criticism, a reaction such as "I'm already doing everything I can" may arise. Both interpretations are valid within that person's own emotional experience and context.

Chapter 5: B3 — Intentional

B3 is the pathway that interprets while generating, retaining, and updating multiple hypotheses about the intent behind an utterance based on observation through SCOPE, and generates a response accordingly.

In B3, the primary reference point is not the literal meaning of the utterance or the receiver's own emotional experience, but rather the purpose behind why the speaker made that utterance. That purpose is not confirmed as a single answer. Instead, it is retained as multiple hypotheses while processing continues. When new information becomes available, the retained hypotheses are updated.

In this pathway, the basis for interpretation is placed in the speaker's communicative purpose. Hypotheses are retained without being confirmed, and responses are generated based on that set of hypotheses.

B3 tends to function effectively in situations where the background and purpose of an utterance are complex. On the other hand, the processing involved in continuously retaining hypotheses may carry a different kind of load compared to pathways that use literal meaning or emotional experience as their basis.

The purpose of B3 is not to identify the correct intent. It is to make the process from interpretation through to response generation observable, while retaining communicative purpose as multiple hypotheses.

[Example] Upon receiving the utterance "Keep it up!", B3 focuses on why this person made this utterance. Hypotheses such as genuinely offering encouragement, wanting to be seen as an ally, applying pressure, or simply saying it out of habit are all retained simultaneously as the response is generated. No single hypothesis is confirmed, and the set of hypotheses is updated as new information becomes available.

Note: SCOPE, referenced in this chapter, will be addressed in detail in a separate paper.

Chapter 6: Examples

The preceding chapters have introduced B1, B2, and B3 individually. Chapter 6 examines what happens when the same utterance is processed through each of the three pathways.

Situation

Someone says "Keep it up!"

Processing via B1

In B1, "Keep it up" is interpreted literally as either encouragement or an instruction. The resulting response is either an assessment of whether further effort is possible, or an expression of gratitude for the encouragement. The situation surrounding the utterance and the relationship with the speaker do not serve as primary reference points.

Processing via B2

In B2, processing proceeds based on how the receiver took in the utterance. The interpretation shifts depending on the receiver's emotional state at the time and their relationship with the speaker. If the words come from someone the receiver feels positively toward, they are likely received as straightforward encouragement and met with gratitude. On the other hand, if the receiver is not in a good place emotionally, or if the speaker is someone they feel negatively toward, the same words may be received as sarcasm or criticism — or may trigger a reaction such as "I'm already doing everything I can."

Processing via B3

In B3, the focus is on why the speaker made this utterance. Hypotheses such as genuinely offering encouragement, wanting to be seen as an ally, applying pressure, or simply saying it out of habit are all retained simultaneously as the response is generated.

Across the three pathways

Faced with the same utterance "Keep it up!", B1 uses the content of the utterance as its basis, B2 uses the receiver's own experience of the utterance, and B3 uses a set of hypotheses about the speaker's communicative purpose. The divergence in interpretation may arise from which pathway is being used.

The B-Series is a model that organizes these differences into an observable structure.

Chapter 7: LLMs and the B-Series

As seen in Chapter 1, divergence in interpretation does not occur only in human-to-human communication.

Many current LLMs may possess processing that references the literal meaning and context of an utterance to generate a response. Within the B-Series framework, this corresponds to processing in the range of B1 through B2.

Regarding the retention of multiple hypotheses about communicative purpose — the domain addressed by B3 — it remains unclear to what extent such processing is functioning.

If this domain is not functioning sufficiently, it may have implications for dialogue settings such as customer support. Even when a response has been carefully designed, if the candidates for communicative purpose are not sufficiently referenced during processing, there is a possibility that the exchange may not function as intended.

The B-Series is a framework for organizing this problem from the perspective of which pathways are functioning. By observing which pathway is being used, it may become easier to identify where divergence in interpretation is likely to arise.

The retention of communicative purpose addressed by B3 will be examined in greater detail in a separate paper.

Chapter 8: Connecting to the COS Series

The B-Series is a model that organizes the process from interpretation through to response generation into three distinct pathways. However, the B-Series does not stand alone as a complete system.

In actual cognitive processing, interpretation functions within a network of interactions with other processing systems. Processes such as decision-making and transmission exist in parallel with the B-Series, or before and after it. COS is a framework that integrates these processing systems together with observation through SCOPE, and the B-Series constitutes one part of that framework.

What the B-Series addresses is the process from interpretation through to response generation. However, how the results of that process are transmitted to the other party falls outside the observational scope of the B-Series.

For example, even if B3 retains multiple hypotheses, the processing involved in how those hypotheses are communicated lies outside the scope of the B-Series. The pathway through which the results of interpretation are conveyed to the other party — that is, the design of output — is handled by a separate processing system.

Within COS, this domain is handled by the C-Series.

The C-Series will be addressed in the next paper.